Designing Sustainability through Proximity: Collaboration in Product-Service Systems

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Abstract

Sustainable value networks in Product-Service Systems (PSS), where firms jointly deliver customer outcomes and share responsibility for sustainable performance, highly depend on collaboration between network actors. Although prior research has identified a range of relational factors that support collaboration, existing studies predominantly examine these factors in isolation. As a result, there is limited understanding of how different relational configurations shape collaboration and sustainability-oriented outcomes in PSS networks.

This study addresses this gap by applying a proximity lens to examine mechanisms that influence collaboration in sustainable PSS models. Building on five established proximity dimensions (geographical, organisational, cognitive, social, institutional), it investigates how different relationship configurations influence collaboration. Accordingly, the research question centres on *how different relationship configurations contribute to sustainability-oriented outcomes in PSS.*

The empirical analysis is based on 5 interviews and the first round of a Delphi study across academia, industry and innovation practitioners experienced in implementing sustainable PSS models. Building on the five dimensions, the resulting blueprint serves as an intermediate step that further supports early-stage PSS design decisions. By clarifying key mechanisms and interdependencies, it shows how design and collaboration choices influence sustainability outcomes and enables comparison of alternative PSS configurations

Moving beyond isolated factors, the paper explains how specific combinations of relational and contextual conditions enable effective collaboration and sustainable value creation. This research advances understanding of how inter-organisational collaboration can be designed and managed in sustainable service provision, offering both conceptual and practical guidance for developing resilient service-based business models.

Keywords: proximity, sustainability, product-service systems

1. Introduction

Sustainability has moved from a normative nice-to-have to an existential imperative for companies: without profound changes to how value is created and distributed in economic systems, environmental degradation, social inequity, and economic instability threaten the foundations of contemporary societies (Stoner, 2011).

Despite widespread recognition of the need for sustainable development, prevailing business models and value creation logics remain largely rooted in linear production and consumption patterns (Nosratabadi *et al.*, 2019). These models insufficiently account for shared responsibility across actors and fail to unlock the full potential for environmental, economic, and social sustainability, the triple-bottom-line.

Product-Service Systems (PSS), in which firms jointly deliver combinations of products and services and share responsibility for performance, have been widely identified as a promising approach to address these challenges (Reim *et al.*, 2015). However, realising this potential requires a shift from firm-centric thinking towards sustainable value networks, in which value is created collaboratively across multiple actors (Gaiardelli *et al.*, 2021). Within such networks, close alignment between manufacturers and consumers is especially critical, as this interface holds significant leverage for sustainability-oriented value creation. Consequently, effective collaboration emerges as a central enabler of sustainable PSS (SPSS).

While prior research has acknowledged the importance of collaboration in PSS, several gaps remain in understanding collaborative structures. Existing studies often focus on dyadic relationships or isolated collaboration factors, offering limited insight into how different relationship configurations jointly shape sustainability outcomes (Li *et al.*, 2020). Furthermore, despite growing attention system collaboration in PSS design, there is still a lack of integrative frameworks that explain the underlying mechanisms through which collaboration emerges, evolves, and is maintained across PSS networks (Zhao *et al.*, 2025). In particular, while proximity has been suggested as a key determinant of inter-organisational collaboration, its multidimensional nature and dynamic interplay in the context of sustainability-oriented PSS (SPSS) remain underexplored.

To address these gaps, this study explores *how different relationship configurations contribute to sustainability-oriented outcomes in Product-Service Systems* by applying a proximity-based theoretical lens. Building on five dimensions, it examines how relational configurations influence collaboration within SPSS value networks. Empirically, the study draws on expert interviews and a Delphi study involving experts from diverse disciplines, industries, and geographic contexts. The Delphi process was designed to introduce the research topic, iteratively build on expert knowledge, and ultimately reach consensus across perspectives.

The study results in a blueprint for collaboration in SPSS that describes the interdependencies among the proximity dimensions and their implications for sustainability outcomes. Rather than providing prescriptive, step-by-step instructions, the blueprint is intended to guide reasoning and decision-making by making underlying relational mechanisms

explicit. This approach is particularly valuable in early design stages, where decision-makers need to explore alternative configurations and assess their potential effects before concrete governance structures are fixed. As such, the blueprint represents an important intermediate step towards more formalised tools, enabling the systematic evaluation and comparison of collaboration strategies in SPSSs. By operationalising this blueprint, the study advances a more generalisable and actionable understanding of collaboration in SPSS.

2. Proximity and Sustainable Product-Service Systems

PSS are widely regarded as a potential vehicle for sustainability due to their systemic integration of products and services across the lifecycle. By shifting from product ownership to outcome-oriented offerings, PSS can improve resource efficiency, extend product lifecycles, and support circular practices through collaboration among network partners (Kjaer *et al.*, 2019; Tukker, 2004). In addition to environmental benefits, PSS can contribute to social sustainability by increasing access to products and enabling new forms of social innovation and inclusion (Spadafora and Rapaccini, 2024).

However, PSS do not inherently guarantee sustainable outcomes. Prior research highlights the risk of rebound effects, where PSS configurations may unintentionally increase consumption and emissions if not carefully designed and managed (Metic and Pigosso, 2022). Realising the sustainability potential of PSS therefore requires deliberate system-level design choices that mitigate such effects in practice. The systemic nature of PSS implies that sustainability outcomes emerge through collaborative action, rather than isolated efforts.

Effective collaboration goes beyond mere cooperation (Bertoni and Larsson, 2011), requiring actors across networks to work toward shared goals, exchange knowledge, and coordinate responsibilities. In this context, the concept of proximity offers a valuable lens to understand how the spatial and relational distances between actors influence collaborative dynamics and innovation performance (Boschma, 2005).

The notion of proximity originated with the French School of Proximity Dynamics and was initially developed in the field of economic geography to explain spatial clustering and regional economic development (Torre and Gilly, 2000). Over time, this perspective has been extended to innovation and knowledge management research, highlighting that different levels and types of proximity can support or constrain knowledge exchange and collaborative learning in networks (Balland *et al.*, 2015). More recently, the proximity framework has been applied to supply chains and multi-actor systems, demonstrating its relevance for understanding how innovation capabilities are shaped by the interactions between connected parties (Bushell *et al.*, 2025; León-Bravo *et al.*, 2025).

Proximity provides a lens to analyse spatio-temporal relational spaces, uncovering the coordination mechanisms that enable actors to work together effectively (Boschma, 2005; Torre and Gilly, 2000). In the literature, proximity is typically divided into two overarching categories: absolute (spatial) proximity and relative (non-spatial) proximity. While geographic or spatial proximity is widely agreed upon and straightforward to measure, definitions of non-spatial dimensions vary, and research continues to refine these concepts.

Building on Boschma's framework, five key dimensions of proximity are particularly relevant for understanding collaboration in PSS networks. Geographical proximity refers to the physical distance between actors, influencing the frequency and ease of interactions, as well as access to innovation hubs and regional clusters. Organisational proximity captures the relational distance between parties in terms of coordination and control mechanisms, such as contracts, agreements, or formalized structures that govern collaborative activities. Cognitive proximity represents the degree of shared knowledge, understanding, and interpretative frameworks between partners, enabling effective communication and joint problem-solving. Social proximity concerns the interpersonal relationships among individuals, fostering trust and close collaboration but also posing risks of favouritism or dependency. Finally, institutional proximity reflects the alignment of values, norms, and cultures at a macro level, including corporate cultures and broader societal practices, which can either facilitate or hinder cooperation across organizational boundaries (Boschma, 2005).

In addition to these five dimensions, further research has highlighted the importance of technological and cultural proximity in innovation systems (Anzola-Román *et al.*, 2019; Streb, 2003). These additional dimensions recognize that shared technical capabilities and cultural understanding can play a decisive role in enabling collaboration and sustaining long-term partnerships across diverse actors.

By examining collaboration through a proximity lens, this study extends the understanding of PSS networks beyond organisational considerations. Proximity provides a structured way to analyse how the relationships, knowledge sharing, and alignment among actors influence the effective implementation of sustainable PSS, offering insights into the mechanisms that underpin long-term, resilient, and innovation-driven collaborations.

3. Research Design

This study adopts a multi-method approach to capture a comprehensive understanding of collaboration within SPSS networks, ensuring diversity in terms of geography, industry, and expertise. By combining expert interviews with a Delphi study conducted via the SoSci Survey platform (Leiner, 2025), the design integrates the depth of qualitative insights with the rigour of structured consensus-building. The expert interviews provide rich, contextualized knowledge that informs the subsequent Delphi rounds, creating a coherent and iterative research process.

Expert selection followed established criteria and combined purposeful and snowball sampling techniques (Hasson and Keeney, 2011). Five initial interviews ensured representation across key domains and regions. An additional 10 interviews will be conducted after the Delphi study to further explore collaborative mechanisms in SPSS. The Delphi study adhered to the RAND approach, emphasizing anonymity, iteration, controlled feedback, and statistical aggregation to support systematic consensus-building and reduce bias (Dalkey and Helmer, 1963).

The initial expert interviews involved experts from academia, industry, and innovation and research centres. All participants received a pre-information package outlining the SPSS concept and relevant proximity dimensions to ensure a shared understanding. These participants

formed the core of the Delphi panel, which was later expanded.

3.1. Topic Exploration – Expert Interviews and Delphi Study

The expert interviews followed a semi-structured format lasting 45-60 minutes. They were designed to explore participants' understanding of SPSS, the role of sustainability, and the significance of collaboration. Interviews also investigated participants' understanding of proximity and its relevance to collaborative processes, with clarifications provided on the specific dimensions considered in this study. The insights gained from these interviews formed the basis for the first round of the Delphi study, which aimed to capture a broad overview of expert perspectives on collaboration in SPSS networks. During this round, experts were invited to reflect on the following four questions:

- Which factors have you experienced as most critical for successful collaboration in SPSS?
- What do you consider the main barriers to collaboration in SPSS?
- Which stakeholders play the most significant role in enabling collaboration within SPSS?
- What strategies or practices have you seen work well to overcome collaboration challenges in SPSS?

3.2.Opinion Assessment and Final Evaluation

Building on the initial interviews, the Delphi study employed iterative rounds designed to refine and consolidate expert knowledge. In the second round, participants will assess the factors identified in the first round, distinguishing between enablers and barriers and explicitly considering the influence of proximity. The third round involves rating these factors on a five- to seven-point Likert scale, enabling quantitative assessment and the identification of consensus across the panel. Where necessary, feedback loops will be used to clarify differing responses, ensuring that the final outcomes reflect a shared understanding and collective agreement among experts.

The final stage of the study synthesizes the Delphi findings into a blueprint for successful collaboration within SPSS networks. This blueprint integrates critical enablers, barriers, and the influence of proximity, providing actionable guidance for both practitioners and researchers. Throughout the process, particular attention was given to the reliability and validity of the results, following established methodological guidelines (Gibbert *et al.*, 2008) to ensure that the findings are robust, replicable, and grounded in a systematic assessment of expert knowledge, as seen in Table 1.

Criterion	**How the criterion is addressed in this study**
Credibility	• Multi-method design combining expert interviews and a Delphi study enabled methodological triangulation. • Experts were selected via purposive and snowball sampling based on predefined criteria. • Iterative Delphi rounds with controlled feedback and anonymity

	reduced dominance bias and strengthened collective judgments.
Transferability	• Experts from academia, industry, and professional practice ensured diversity across the innovation ecosystem. • Broad geographic and industry coverage increased contextual variation. • The study context (collaboration in manufacturing SPSS networks) is clearly defined to support analytical generalization.
Dependability	• A structured research protocol was applied across interviews and Delphi rounds. • Semi-structured interview guidelines and a predefined three-round Delphi process ensured procedural consistency.
Confirmability	• Anonymity and statistical group responses in the Delphi process minimized individual and researcher bias. • Findings emerged from iterative expert consensus rather than single interpretations.

Table 1: Description of Rigour in the Methodological Approach in Accordance with (Gibbert et al., 2008)

4. Findings

The presented findings are based on the results of Round 1 of the Delphi study as well as the first five conducted expert interviews. After the final analysis of this round, Round 2 will be designed by generalising the answers according to the theoretical proximity lens and focusing on factors specifically relevant for collaboration in SPSS. Further, the first version of the blueprint based on Round 1 will be presented to the panel to be tested for its validity.

4.1.Factors for Successful Collaboration in SPSS

In the first question of Round 1, participants were asked to identify the most critical factors for successful collaboration in the development and implementation of SPSS. Overall, the responses reveal a strong emphasis on organisational aspects, complemented by technological and strategic considerations. Across the dataset, successful collaboration is framed as a multi-dimensional challenge that requires alignment at strategic, operational, and individual levels, as well as a shared commitment to value co-creation.

A central theme emerging from the responses is the importance of organisational alignment and formal governance structures. Participants frequently highlighted the need for clear contractual agreements, such as NDAs, service-level agreements (SLAs), and early clarification of intellectual property ownership. These mechanisms are seen as essential for establishing transparency, reducing uncertainty, and preventing conflicts later in the collaboration. In particular, agreeing on principles for risk and revenue sharing was identified as critical, as SPSS often expose both providers and customers to operational and financial risks. Well-defined and

mutually beneficial agreements are therefore considered a foundation for trust and long-term collaboration.

Beyond formal structures, respondents consistently stressed the importance of shared or complementary understandings of value, service, and strategies. Successful collaborations are characterised by alignment around value propositions that benefit all stakeholders across the value chain, rather than privileging a single actor, the OEM in most cases. This includes a shared understanding of the joint goal of the SPSS, the intended customer value, and the broader sustainability objectives. Several participants emphasised that such alignment should be supported by explicit documentation and, where possible, simulation models that help partners develop a common understanding of how the SPSS will operate in practice. In addition, co-location and context-specific adaptation were noted as relevant factors, particularly when SPSS are implemented across different markets or regions.

Closely related to this is the need for early and inclusive collaboration around problem solution definition. Participants noted that involving partners early in identifying and validating needs creates a sentiment of shared ownership throughout the lifecycle of the SPSS, from design, to implementation, operation and delivery. This approach contrasts with more traditional supplier and customer relationships and is considered essential when the aim is genuine collaboration and co-creation across the value chain.

Another recurring theme concerns organisational mindset and internal alignment. Participants pointed out that collaboration in SPSS requires acceptance of mutual dependence, particularly at the management level. At the same time, internal buy-in across all organisational levels is crucial: employees need to understand not only their own roles, but also the roles of partners within the wider ecosystem. A shared language, clear responsibilities, and consistent design principles were mentioned as enablers of such alignment.

From a technological perspective, technological maturity and data-related capabilities were identified as important enablers of collaboration. Willingness to share data, supported by shared data standards, platforms, and APIs, was frequently mentioned. Effective data and information sharing are seen as prerequisites for coordination, performance monitoring, and continuous improvement of SPSS.

Finally, several respondents highlighted the importance of value co-creation capabilities and interdisciplinary perspectives. Designing value propositions that account for stakeholder diversity, and integrating technological, business, consumer, policy, and environmental perspectives, were considered essential for addressing the complexity of SPSS. This includes carefully defining system boundaries, for example in lifecycle assessments, to ensure that sustainability impacts are meaningfully captured and aligned with collaborative goals.

4.2.Main Barriers for Collaboration in SPSS

When asked to reflect on the main barriers to collaboration in SPSS, participants described a wide range of interrelated challenges spanning organisational, cultural, strategic, technological, and institutional dimensions. Overall, the findings indicate that many barriers stem from misalignment, whether in power relations, strategic intent, organisational mindsets,

or technical infrastructures, combined with the inherent complexity and uncertainty of SPSS.

A prominent barrier concerns misaligned incentives, power imbalances, and unclear value distribution. Several participants noted that collaboration becomes difficult when individual gains or advantages are not clearly articulated, leading actors to question the rationale for engagement. Power asymmetries between partners can exacerbate this issue, particularly when dominant actors shape the collaboration in ways that undermine perceived fairness or mutual benefit. In such cases, the absence of a clearly established win–win business case, especially for customers transitioning from product ownership to service-based offerings, was identified as a major obstacle.

Closely related to this is the challenge of divergent understandings of service, value, and success. Participants highlighted that partners often hold different perceptions of what constitutes a service and how value should be created and measured. This divergence extends to performance metrics, where traditional indicators such as profitability or return on investment may conflict with sustainability-oriented goals related to environmental impact, lifecycle performance, or emissions reduction. The lack of shared, meaningful metrics for evaluating both economic and sustainability outcomes can hinder communication, slow decision-making, and weaken commitment to the collaboration.

Organisational and cultural barriers were also strongly emphasised. Many respondents pointed to internal processes, structures, and cultures within organisations, particularly manufacturing firms, that remain optimised for product sales, transactional relationships, and efficiency-driven risk mitigation. Such organisations often struggle to adopt relational, service-oriented, and long-term perspectives required for SPSS. Decision-making bodies and senior leadership were described as anchored in product-centric thinking, sometimes reinforced by groupthink or a limited willingness to engage in long-term, exploratory strategies. Cultural differences between collaborating organisations, as well as internally unclear roles and core competencies, further contribute to role conflicts and coordination difficulties, an illustrative example of a rather destructive interplay between institutional and organisational proximity.

Another significant set of barriers relates to trust, commitment, and collaboration dynamics. Participants noted that superficial commitment to co-created SPSS frequently leads to project breakdowns. A lack of trust can discourage openness and data sharing, as firms fear that sensitive information may be misused to their disadvantage. Missing or weak joint visions regarding sustainability goals and strategic intent further undermine collaboration, as partners may pursue fundamentally different outcomes, such as cost reduction, reputation enhancement, or market expansion.

From a technological perspective, system complexity and incompatibility were repeatedly identified as major barriers. Limited understanding of the full value or service chain, coupled with interoperability issues and incompatible interfaces, makes coordination across organisational boundaries difficult. Challenges related to data sharing were also highlighted, specifically regulatory constraints or insufficient technological maturity. Inadequate shared platforms, information systems, and communication environments can significantly impede effective collaboration and transparency.

Finally, participants pointed to external and contextual constraints, including regulatory differences across regions. Variations in regulations governing service provision, data handling, and SPSS management can complicate collaboration, particularly in cross-border contexts. More broadly, uncertainty regarding responsibilities, performance outcomes, and long-term viability increases perceived risk and reinforces organisational reluctance to commit resources to collaborative SPSS initiatives.

4.3.Significant Stakeholders

In response to the third question, participants were asked to identify which stakeholders play the most significant role in enabling collaboration within SPSS. The answers indicate that collaboration in SPSS is inherently multi-actor and ecosystem-based, with different stakeholders assuming varying degrees of influence depending on the market context, system configuration, and maturity of the offering. Nevertheless, several stakeholder groups consistently emerged as particularly influential in enabling or constraining collaborative dynamics.

Across responses, manufacturers and service providers were most frequently identified as central enablers of collaboration. As system owners or integrators, manufacturers typically shape the technical architecture of the SPSS, define the core value proposition, and coordinate participation across the ecosystem. Their ability to deliver SPSS offerings and clearly communicate their value, both internally and externally, was seen as critical. In many cases, manufacturers and service providers were described as needing to operate in very close alignment, or even as a single integrated entity, to effectively manage the transition from product-centric to service-oriented business models.

At the same time, participants highlighted that the customer plays a pivotal role in determining the viability and direction of SPSS collaboration. Customers ultimately define the core value proposition by expressing needs, setting performance expectations, and deciding whether to adopt service-based offerings. Several respondents noted that large or strategically important customers, in particular, possess significant power to influence manufacturers' willingness to shift from traditional product sales to SPSS models. Without customer understanding, acceptance, and demand for SPSS, collaborative efforts are unlikely to succeed, regardless of technological or organisational readiness.

Policymakers and regulators were repeatedly mentioned as influential enablers next to service providers, particularly in relation to sustainability objectives. Participants generally agreed that sustainability-driven transitions toward SPSS are often externally motivated rather than intrinsically driven by market actors. Well-designed policies, regulations, and funding mechanisms can therefore play a key role in accelerating SPSS diffusion by creating incentives, setting standards, and lowering adoption barriers. However, respondents also cautioned that policy interventions require a deep understanding of ecosystems and value chains, as poorly designed regulations may introduce administrative burdens or inadvertently disadvantage smaller organisations.

Further, service delivery actors, intermediaries and value-chain partners, such as operators,

dealers, and other service providers, were identified as critical in certain contexts. In markets where manufacturers do not interact directly with end customers, dealer networks and intermediaries act as essential interfaces for maintenance, service delivery, and customer engagement. Participants emphasised that excluding such actors from SPSS development and value streams can severely limit scalability and adoption, particularly in traditional or conservative market segments.

In the same vein, several participants further pointed specifically to the role of technology service providers in enabling collaboration by supporting digitalisation, data sharing, and interoperability. Providers of shared platforms, communication infrastructure, and data services were seen as instrumental in creating the technical conditions for effective collaboration, often through solutions that manufacturers can integrate into their SPSS offerings.

Lastly, an emerging theme in the responses was the potential value of independent third-party orchestrators or system designers. In contrast to ecosystems dominated by a single OEM, such neutral actors can facilitate collaboration by aligning incentives, mediating conflicts, and ensuring balanced participation among stakeholders. Participants suggested that while this ideal of equality among actors is rarely realised in practice, independent orchestration may help overcome power asymmetries and rigid supply-chain logics that hinder collaborative SPSS development.

4.4.Strategies to Overcome Collaboration Challenges

In the final question, participants were asked to reflect on strategies and practices that have proven effective in overcoming collaboration challenges in SPSS. The responses suggest that successful strategies focus less on isolated tools or technologies and more on establishing clear structures, fostering trust-based relationships, and reinforcing shared strategic intent over time. Effective collaboration is thus described as an ongoing, actively managed process rather than a one-off organisational arrangement.

A recurring theme across the responses is the importance of early and explicit alignment on goals, roles, and expectations. Participants consistently emphasised the need to clearly define objectives, requirements, performance expectations, responsibilities, and timelines at the outset of collaboration. This early alignment, particularly at the management level, helps prevent later misunderstandings and provides a shared reference point when challenges arise. Ensuring that the collaboration is perceived as a win–win for all parties was identified as essential for sustaining long-term commitment.

Structured and especially meaningful communication practices emerged as another central strategy. Regular meetings were widely regarded as necessary, but participants stressed that their effectiveness depends on clear agendas, defined ownership, and genuine commitment from participants. A combination of frequent online meetings and targeted in-person workshops was seen as particularly effective, especially in early phases where a shared understanding and team spirit need to be established. Occasional involvement of senior or executive-level leaders in these interactions was highlighted as a way to reinforce the strategic importance of the collaboration and signal long-term commitment.

Closely related to communication is the role of relationship-building and trust development. Face-to-face interactions, such as on-site visits, co-location over extended periods, staff exchanges, or joint workshops, were repeatedly identified as highly effective, albeit underutilised, practices. These activities help build mutual respect, deepen understanding of each other's constraints and capabilities, and foster trust at both individual and organisational levels. While informal social activities were mentioned as supportive, participants generally emphasised that trust is primarily built through open, transparent discussions about project challenges, risks, and trade-offs.

From an organisational and governance perspective, clear collaboration rules and leadership commitment were seen as critical enablers. Well-defined contractual frameworks, including agreements on data sharing, intellectual ownership, and confidentiality, were considered prerequisites for openness and transparency during collaboration. Participants also pointed to the importance of leadership, either through a clearly defined coordinating actor or a fair and trusted system leader, who can hold the collaboration together, resolve conflicts, and ensure that no partner is treated unfairly. Leadership attitudes and behaviours were described as particularly influential in shaping employee perceptions and willingness to engage openly.

Transparency and shared working infrastructures were further highlighted as practical strategies for overcoming collaboration challenges. Shared documentation, common information platforms, and open access to meeting minutes, simulations, calculations, and customer insights were described as critical for maintaining alignment and enabling joint problem-solving. Common customer meetings and interviews were also seen as effective in aligning heterogeneous teams around user needs and service expectations, particularly when partners have different backgrounds or levels of customer access.

Finally, participants emphasised the importance of long-term strategic orientation. Rather than focusing solely on short-term business models or pilot projects, effective collaborations were described as guided by jointly developed roadmaps for long-term SPSS operation and continuous improvement and iterative processes. Such roadmaps clarify strategic priorities by helping partners navigate uncertainty and adapt collaboratively over time.

4.5.Blueprint and Next Steps

While Round 1 identified key factors, barriers, stakeholders, and strategies, these findings cut across multiple dimensions of proximity without explicitly distinguishing between them and many of the answers can be attributed to collaboration in general. Therefore, building on the exploratory insights gathered in Round 1, the second round of the Delphi study aims to structure, contextualise, and deepen participants' reflections by introducing the proximity perspective and focusing on aspects specifically relevant for SPSS. Round 2 thus seeks to map the Round 1 findings to specific proximity dimensions, assess the perceived importance and role of each dimension in enabling or constraining SPSS collaboration, and prompt judgement on how different forms of proximity influence and interact across SPSS contexts. For this, participants will be confronted with the first blueprint version, as seen in Figure 1.

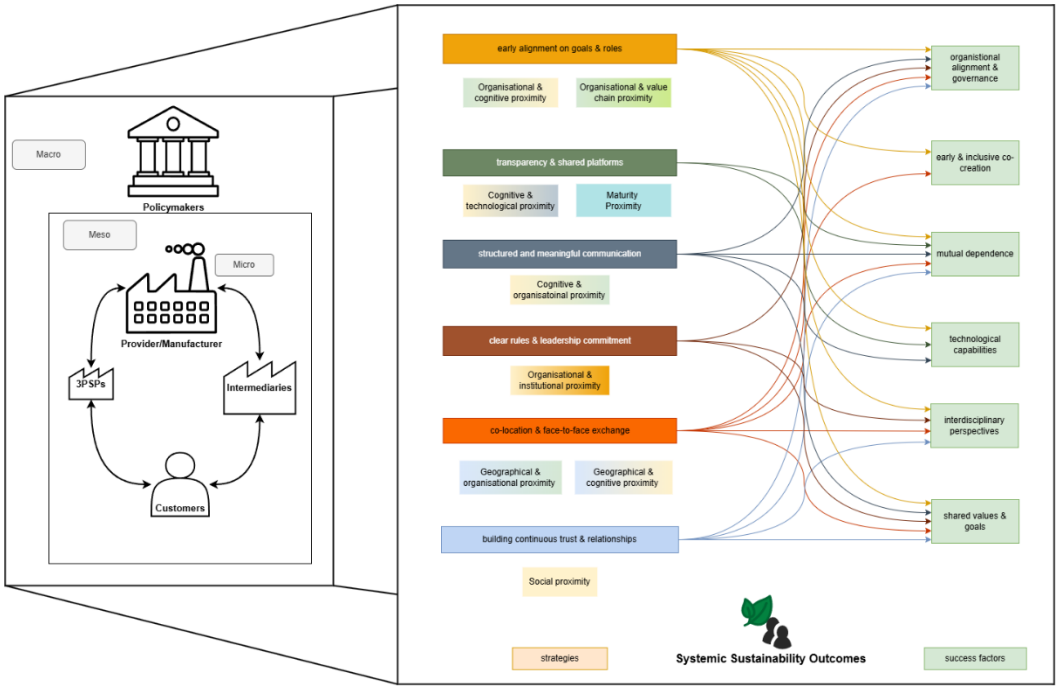


Figure 1: Intermediate Blueprint after Round 1. Size of Actor Nodes Indicates Number of Mentions.

Integrating the findings of the first expert interviews with Round 1, the proximity dimensions considered in Round 2 are geographical, organisational, cognitive, social, institutional, technological, and value chain proximity (i.e. relative position and role within the value chain).

5. Discussion

One of the key insights emerging from the findings is the need to extend existing proximity frameworks to better capture the realities of SPSS collaboration. While organisational, cognitive, social, institutional, and geographic proximity were all reflected in the participants' responses, two additional or reframed dimensions appear particularly significant.

First, the findings point to the relevance of what can be described as maturity proximity. Across the responses, participants repeatedly referred to technological maturity, organisational readiness, service capabilities, and experience with SPSS as critical enablers or barriers to collaboration. These aspects are not fully captured by existing proximity dimensions, yet differences in maturity between partners can strongly affect power relations, trust, expectations, and the ability to jointly develop and scale SPSS. Maturity proximity therefore captures the extent to which actors are aligned in their readiness to engage in complex, service-oriented, and sustainability-driven innovation.

Second, the notion of value chain proximity, sometimes expressed through references to intermediaries, system integrators, or ecosystem roles, emerged as a distinct and influential dimension. It has previously been coined as supply chain proximity (Cortes, 2023).

Participants' emphasis on manufacturers, service providers, and intermediaries suggests that collaboration challenges in SPSS are not only about distance or similarity between organisations, but also about their relative positions, dependencies, and roles within the value chain. This role-based or intermediary proximity shapes access to customers, control over data and interfaces, and the ability to influence system design, making it a critical lens for understanding collaboration dynamics beyond dyadic relationships.

The findings also underscore the importance of technological proximity as a distinct dimension that deserves more explicit attention in SPSS research. While cognitive proximity partially addresses shared knowledge and understanding, participants' repeated references to interoperability, digital platforms, APIs, and system compatibility suggest that technological proximity encompasses more than shared cognition. The technological complexity and digital transformation inherent in many SPSS require alignment not only in understanding, but also in technical architectures, standards, and capabilities.

This distinction becomes particularly important given that SPSS often rely on digital infrastructures to enable service delivery, monitoring, and continuous improvement. Insufficient technological proximity can therefore undermine collaboration even when strategic intent and values are aligned. Introducing technological proximity as a separate analytical category allows for a more precise understanding of how digitalisation both enables and constrains collaborative SPSS development, further aligning with research on digital servitization (Beducci *et al.*, 2025).

A notable tension in the findings concerns the apparent dichotomy between critical collaboration factors and effective strategies. In Round 1, participants primarily identified structural and organisational factors as critical for successful collaboration. In contrast, when discussing strategies to overcome collaboration challenges, respondents emphasised relationship-building practices, including trust development, face-to-face interactions, co-location, and informal exchanges.

This discrepancy suggests that relationship-building may be perceived less as a foundational success factor and more as a remedial or enabling mechanism that becomes apparent when collaboration is strained or when structural arrangements alone prove insufficient. One interpretation is that experts take relational aspects such as trust and mutual understanding for granted at the beginning, only recognising their importance once collaboration difficulties arise. Alternatively, this dichotomy may reflect the inherent uncertainty and complexity of SPSS, where formal structures are necessary but not sufficient, and relational aspects become critical in navigating unforeseen challenges.

From a proximity perspective, this finding highlights the dynamic interplay between organisational, social, and cognitive proximity. While organisational proximity can be deliberately designed through contracts and governance, social and cognitive proximity often need to be actively cultivated over time. The results therefore suggest that effective SPSS collaboration requires not only the establishment of formal governance, but also ongoing investment in relational aspects as the system matures.

6. Conclusion

This study examined collaboration dynamics in SPSS through expert interviews and a Delphi study, aiming to identify key factors, barriers, stakeholders, and strategies for sustainability-oriented collaboration. Applying a proximity lens enabled a nuanced understanding of how spatial and relational proximity dimensions shape collaborative outcomes. The findings highlight both established and underexplored aspects of proximity in SPSS and reveal conceptual tensions in how collaborative relationships are configured to support sustainability goals.

The results advance theory by demonstrating that different proximity dimensions become salient for SPSS design. The introduction of maturity, value chain and technological proximity reflects the complexity of digitally enabled sustainability-oriented systems. Beyond SPSS, these insights contribute to research on business model innovation and sustainable business models by positioning proximity and collaborative configurations as critical mechanisms for achieving sustainability outcomes.

From a practical perspective, the findings indicate that effective SPSS collaboration extends beyond partner selection and contractual arrangements. Organisations must align maturity levels, clarify ecosystem roles, ensure technological compatibility, and actively cultivate relational proximity. Integrating structural alignment with ongoing relationship-building can help transform collaborations into more resilient and adaptive SPSS ecosystems.

Despite the study's contributions, several limitations warrant attention. The sample size and composition of the Delphi panel may constrain the generalizability of the findings, particularly across industries or geographic contexts not represented. Furthermore, while the research aims to provide a conceptual blueprint for collaboration, simulating collaborative processes and operationalizing the identified proximity dimensions in practice remains a challenge. Future research could explore how manipulating different proximity dimensions influences collaboration effectiveness and sustainability outcomes in SPSS, potentially through simulation studies or longitudinal empirical designs.

This study underscores the need for a more nuanced and dynamic application of the proximity perspective in SPSS research. Collaboration is shaped by an evolving balance between formal structures, relational practices, and emerging dimensions of proximity. A deeper understanding of these dynamics offers both researchers and practitioners actionable guidance for designing and managing collaborations that are resilient and aligned with sustainability goals.

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